

BTEC Level 3 Applied Science · Unit 1 Biology · Worksheet**Cell Structure**

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. State the function of the nucleus, a mitochondrion and a ribosome.
2. Name three structures found in plant cells but not in typical animal cells.
3. What is meant by a selectively permeable membrane?

B Knowledge

4. Describe the structure of the cell-surface membrane.
5. Explain how cristae help mitochondria to transfer energy.
6. Describe the role of the cellulose cell wall.

C Application

7. Why do muscle cells contain many mitochondria?
8. Root cells usually lack chloroplasts. Explain why this is not a problem for the plant.
9. Explain why mature mammalian red blood cells cannot produce new proteins.

D Exam-style practice

10. Describe the functions of the nucleus and mitochondria in an animal cell. [4]
11. Explain two ways a palisade cell is adapted for photosynthesis. [4]
12. Distinguish between the cell wall and the cell membrane. [3]

E Challenge

13. Secretory cells have extensive RER and Golgi bodies. Explain this combination in terms of protein trafficking.